PRIVATE RESEARCH - NOT INVESTMENT ADVICE

The funding-risk engine, unified scientific report

final edition - 8 November 2026 - prepared for Tijs de Boer

The short version

1. All five answers in one line each: (1) the utility formula barely matters - CARA, CRRA, HARA and the kinked objective pick the same mix within ~0.5% of score; (2) rebalance rarely - annual wins once trading costs are charged, by a small margin; (3) horizon is the biggest lever here - longer horizons need much larger T-bill shares, and just holding the 10-year mix fails the regime law at 20+ years; (4) the in-period cap is cheap until it bites - at your 40% it costs nothing, at 5-10% it dominates everything; (5) clever has still not beaten simple.
2. Everything new in this report was run on a fresh seed family (20261109), disjoint from all earlier studies, and judged under both market laws and both carry views. Where an older seed family says something different, both are shown.

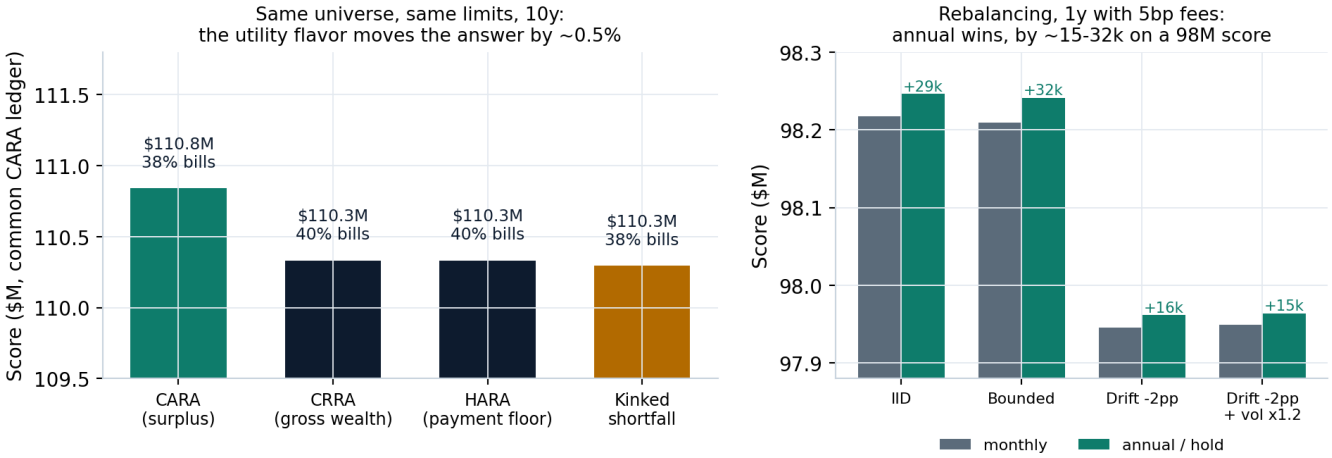
Inside: 1. Utility models - 2. Rebalancing frequency - 3. Investment horizon - 4. The in-period limit - 5. Clever vs simple - 6. What is most worth telling the group - Appendix: evidence and provenance

1. Utility models: CARA, HARA and the others barely move the answer

You asked how the choice of utility model - CARA, CRRA, HARA, or the kinked-shortfall objective - changes the outcome. Short answer: at your $250k/month it does not, to any degree that matters.

Objective	What it rewards	Bills + anchor	Score (10y, new model)
CARA (surplus)	steady dollar aversion to missing the surplus target	38%	$110.8M
CRRA (gross wealth)	relative aversion on total wealth	40%	$110.3M
HARA (payment floor)	aversion that explodes near the payment floor	40%	$110.3M
Kinked shortfall	5:1 extra penalty on falling short	38%	$110.3M

- All scores are re-expressed on one common scale (the CARA cash-first ledger), so the table compares like with like. The spread between the four is about $0.5M on a $110M score.
- In the one-year study (identical paths, all four objectives, all-risk judge) the four picked the exact same portfolio. The reason is structural: your hard limits - fund every payment, 40% worst-path cap, 15% terminal cap - bind long before the utility's appetite for risk gets a vote. The constraints, not the utility, set the safe share.
- The flavor only starts to matter in the stress fixtures where the plan is close to infeasible (e.g. $95M due at horizon: nobody has a valid candidate, and HARA's score is mathematically undefined below the floor). That is where a utility choice becomes a real decision - not at $250k/month.
- Payout size moves the safe share far more than the utility does: at $50k/$100k/$250k per month the validated mix is nearly unchanged because the loss caps, not the payments, bind.



Left: four objectives, same universe and limits, 10y - the scores sit within ~0.5% and the safe shares within 2.5 points. Right: rebalancing at 1y with fees, per market law; annual leads by a few tens of thousands of dollars.

2. Rebalancing frequency: rarely wins, and the prize is small

Monthly, quarterly, semiannual and annual rebalancing were compared on identical paths, same starting mix, same payments, with 5bp trading fees and one-week settlement - the realistic friction case.

Market law	Monthly	Quarterly	Semiannual	Annual / hold
IID nominal	98.217	98.224	98.230	98.246
Bounded nominal	98.209	98.216	98.224	98.241
Equity drift -2pp/yr	97.946	97.950	97.953	97.962
Drift -2pp, vol x1.2	97.949	97.953	97.956	97.964

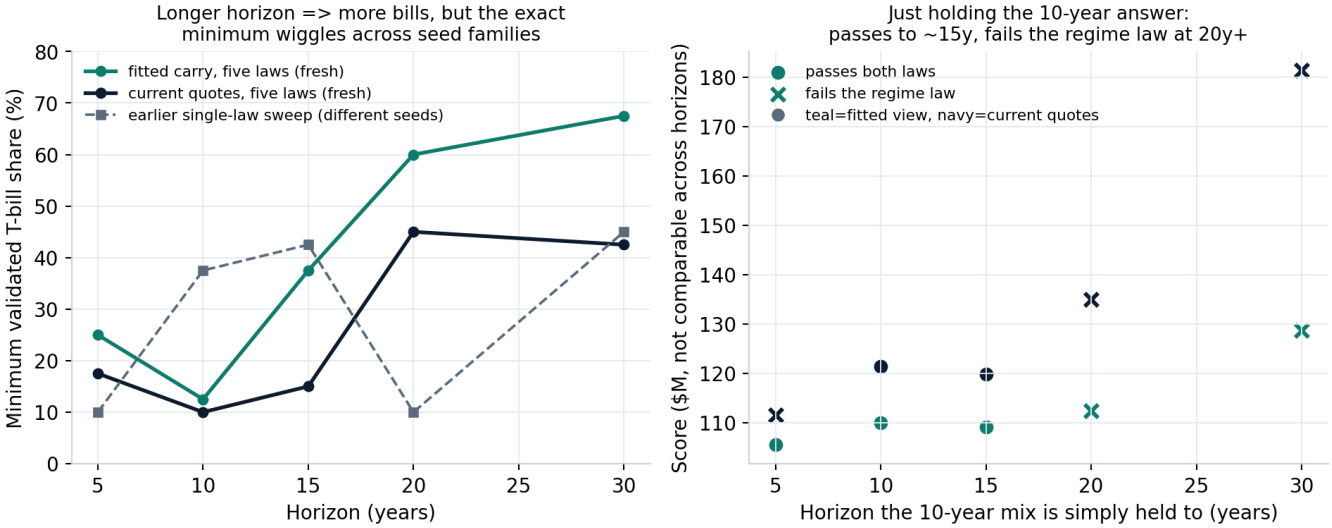
- Scores are certainty equivalents in $M over the one-year test window; all 16 rows pass every risk and funding check. Annual wins in all four laws - by $15-32k, i.e. hundredths of a percent. Fees explain the ordering: monthly trading costs ~$1.7k/yr here, annual ~$0.

- At 3 and 5 years, no frequency passes all four laws under the unchanged budgets - reported, not hidden. Annual still ranks first by score ($54-174k over monthly), but that is a ranking of tested schedules, not a certified optimum.
- A fee-free 10y twin study (no buy-side cost, labeled) flips the sign: monthly scores marginally higher ($111.38M vs $110.84M buy-and-hold). So the honest rule is: at real fee levels, rebalance about annually; the decision matters less than every other choice in this report.
- Cleverer does not help: the learned feedback policy that watches funding, prices and momentum loses to its own first-trade-only twin in attribution, and the robust four-law search found no acceptable candidate at all.

3. Investment horizon: the biggest lever of the five

Same $250k/month, same limits (the 15% terminal cap sits at the end of each horizon, as you specified), horizons of 5, 10, 15, 20 and 30 years. New this report: the whole sweep re-run on a fresh seed family under both carry views and both judging laws.

Horizon	Earlier sweep (single law)	Fresh: fitted view	Fresh: current quotes	Fresh: current quotes score	Holding the 10y mix works?
5y	10%	25%	18%	$111.9M	NO
10y	38%	12%	10%	$121.4M	yes
15y	42%	38%	15%	$122.2M	yes
20y	10%	60%	45%	$145.1M	NO
30y	45%	68%	42%	$187.4M	NO



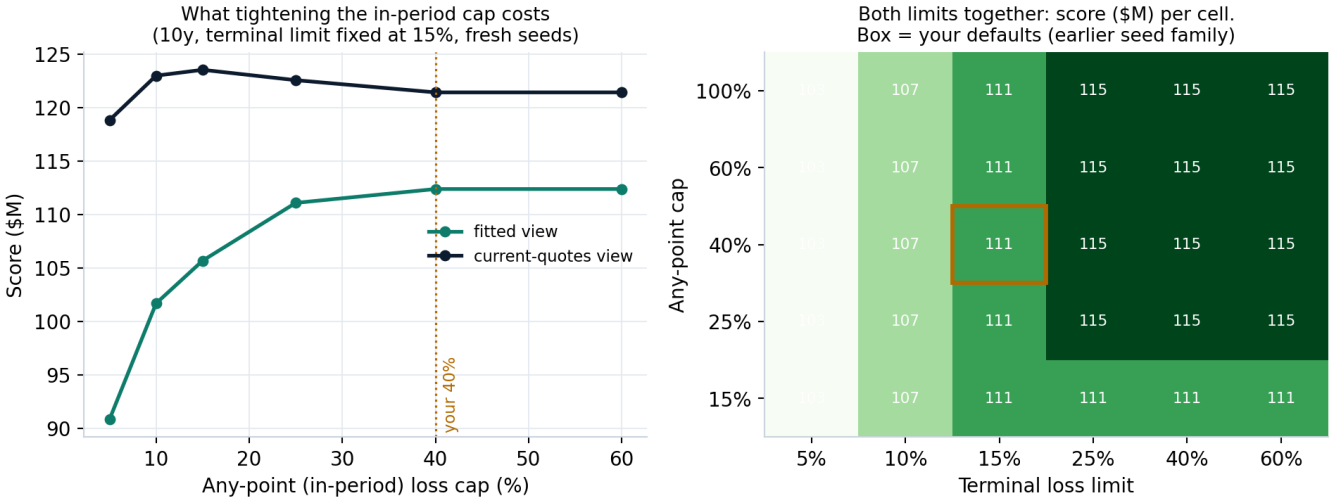
Left: minimum validated T-bill share by horizon - the trend is steeply up, the wiggles are seed noise at a flat frontier. Right: the frozen 10-year mix simply held to other horizons - fine to about 15 years, rejected by the regime law at 20+.

- The trend is the answer: the longer the money must last, the larger the safe share the judge demands - from ~10-25% bills at 5 years to ~45-68% at 20-30 years. More time is more chances for a deep drawdown before the last payment, and the 15% terminal cap has to survive all of them.
- The wiggles are honest noise: at 10y the fresh fitted run validates at 12.5% bills where the round-2 seed family needed 32.5%. The score difference between those cells is small; the frontier is flat, so the exact minimum is seed-sensitive. Do not read precision into any one cell - read the slope.
- CE levels across horizons are NOT comparable: 30 years pays out $90M of the $100M, 5 years pays $15M. The rising scores at long horizons mostly reflect more years of compounding on what is left, not a better deal.
- Practical: the current 10y answer is robust for horizons up to about 15 years. If the real mandate is 20+ years, the mix must be re-solved for that horizon - holding the 10y mix fails the regime law's terminal-loss check at 20 and 30 years on fresh paths.

4. The in-period limit: free until it bites, then it owns the portfolio

The in-period (any-point) cap limits how far underwater the plan may go at any moment, not just at the end. Your default is 40% with a 15% terminal limit. Fresh 10y sweep, both views, terminal limit fixed at 15%:

Any-point cap	Fitted: bills / score	Fitted: below best cell	Quotes: bills / score	Quotes: below best cell
5%	88% / $90.8M	$21.5M	75% / $118.8M	$4.7M
10%	65% / $101.7M	$10.7M	52% / $123.0M	$0.5M
15%	52% / $105.7M	$6.7M	40% / $123.5M	$0.0M
25%	25% / $111.1M	$1.3M	20% / $122.6M	$1.0M
40%	12% / $112.4M	$0.0M	10% / $121.4M	$2.1M
60%	12% / $112.4M	$0.0M	10% / $121.4M	$2.1M



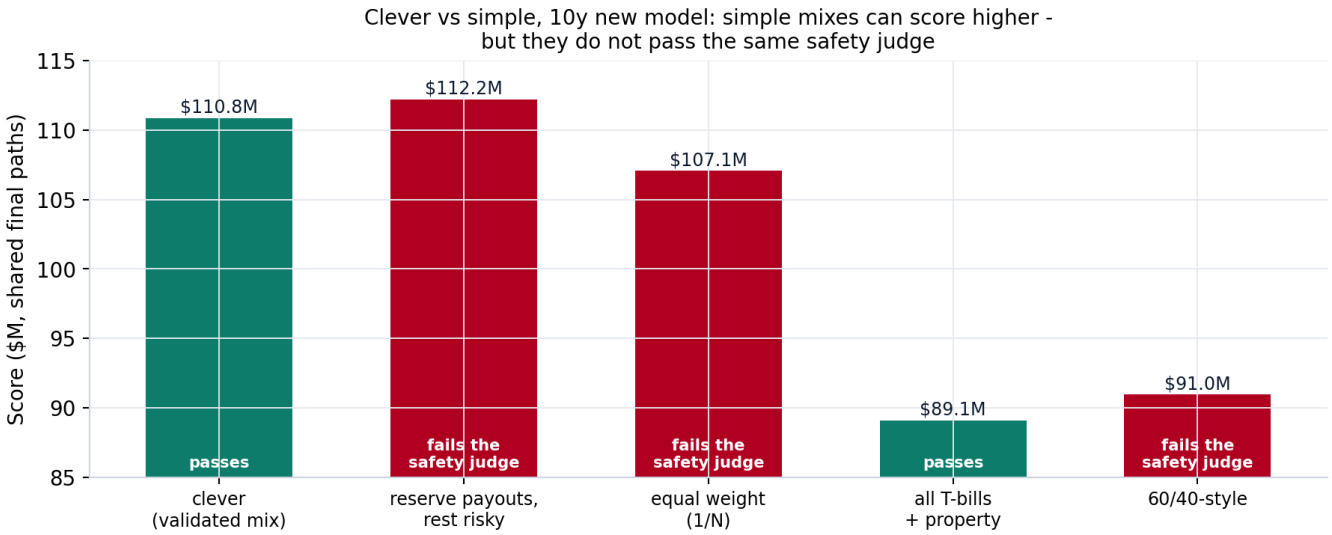
Left: score vs the any-point cap at a fixed 15% terminal limit (fresh seeds). Right: the full two-limit grid from the earlier seed family - the box marks your 40% / 15% defaults.

- At your 40% the cap is slack: loosening it to 60% changes nothing, because the 15% terminal limit binds first. Tightening to 25% costs the fitted view about $1.3M; to 15%, about $6.7M; to 5%, over $21M and 87.5% of the portfolio in bills.
- Under current quotes the same pattern with smaller costs (bill carry does more of the work): the 15% in-period cap even scores slightly above the 40% cell - a flat-frontier wiggle, not a free lunch; the 5-10% caps still cost $3-5M.
- So the choice is genuinely yours: the cap is cheap insurance at 25-40% and very expensive below ~15%. If the point of the cap is 'never show the client a scary number', 15% costs roughly $4-7M of score; 40% costs nothing but allows deep interim drawdowns.

5. Clever vs simple: simple has not been beaten

Benchmarking the validated mix against the plain alternatives on identical paths, under the same safety judge (10y new model):

Candidate	Score	Safety judge
Clever: the validated mix	$110.8M	passes
Reserve payouts, rest risky	$112.2M	FAILS validation
Equal weight (1/N)	$107.1M	FAILS validation
60/40-style	$91.0M	FAILS validation
All T-bills + property	$89.1M	passes



Green passes the full judge, red fails it. Two simple mixes score above the validated mix - and neither survives the safety checks that produced it.

- The honest nuance: 'reserve payouts, rest risky' scores $1.4M MORE than the validated mix - on paths it was then rejected by. Scoring higher while failing the judge is not winning; the judge is the point.
- The historical record agrees: 19 frozen annual holdouts of real history, matched objectives, 152 outer refits - no objective formula, forecast model or trading rule beat the plain fixed mix once safety was enforced.
- The frozen policy-by-model matrix (5th edition): the two 1y-sleeve mixes and the two 10y-sleeve mixes sit within a few $M of each other under every law, including the regime law and real terms. No policy wins everywhere; complexity never wins anywhere.
- The clairvoyant relaxation (a cheater who sees next week's returns) bounds any policy at $197-253M - far above everything tested. There is no near-optimality certificate; what we can say is narrower and stronger: nothing implementable that we tested beat the simple mix.

6. Most worth telling the group

- The inflation illusion: the recommended mix scores $120.7M nominal but $79.4M in today's dollars. No fixed mix passes a 15% real final-loss limit; the tightest real limits are about 45% worst-path / 40% final. Nominal safety is partly inflation.
- Two honest answers beat one fake one: fitted history says hold 32% bills ($109.2M), today's quotes say 10% ($120.7M), because bills pay 4.17% now vs ~1.8% in the shrunk history. The engine shows both and does not average them.
- Flat optima are treacherous: the true 10y solve reallocates most of the risky sleeve between seeds while the score moves ~$1M. An optimizer that swings wildly is telling you the objective is flat, not that the data changed.
- The model-wrongness margin is a budget, not a calibration: a statistically real Wasserstein radius is ~100,000x the placeholder and unaffordable under the payment budgets. Choose the budget, then the margin - not the reverse.
- Hard limits dominate preferences: with a 99% funding rule and loss caps, the utility function is nearly irrelevant. Groups often spend weeks on the objective and minutes on the constraints; here the constraints did all the work.
- Data rights note for sharing: the histories are Yahoo-sourced for private research; the report is fine to discuss, the data bundle is not cleared for redistribution.

Appendix. Evidence and provenance

- New in this report (seed family 20261109, disjoint from all earlier studies): horizon sweep 5/10/15/20/30y and the in-period cap slice, both carry views, dual-law judging (studies/final5/results/).
- Carried in unchanged: utility comparison (7 Nov, 1y + 10y new-model), rebalancing frequency (1y/3y/5y with fees; 10y fee-free twin), benchmarks and the new-model cml sweep (seed 20261107), round-2 dual-law mixes, true-10y solves, real-terms repricing (seed 20261108).
- Score definition: certainty equivalent - the guaranteed dollar amount after all payments worth exactly as much as the risky plan, expressed on the common CARA cash-first ledger.
- Judge: frozen mixes on independent paths under the mixture law and the EWMA regime/bootstrap law; funding (99% all-payments), per-payment tails within the 20% allowance, 40% any-point and 15% terminal loss caps. Passing is a statement about the model, never a guarantee.
- Every results file is SHA-256-stamped in studies/RUN_MANIFEST.json (code commit, configs, data, scripts, package versions). The refactor plan for the shared accounting engine is docs/REFACTOR_PLAN_SHARED_ACCOUNTING.md - not started, as agreed.
- Private research for Tijs de Boer. Placeholder client ($100M, $250k/month); not investment advice; only 12 months of payments are owner-confirmed.